

The Problem

We are given:

1. A matrix A , which is a 2×3 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}.$$

This means:

- A has 2 rows and 3 columns.
- Each element a_{ij} is just a number, where i is the row index and j is the column index.

2. A function:

$$f(A) = K = A^T A,$$

where A^T is the **transpose** of A , and $A^T A$ is the multiplication of A^T and A .

We need to calculate the **gradient of K with respect to A** .

Step 1: What Is A^T ?

The transpose of a matrix, A^T , flips its rows and columns. For A , the transpose A^T is:

$$A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}.$$

Notice:

- The first row of A becomes the first column of A^T .
 - The second row of A becomes the second column of A^T .
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Step 2: What Is $A^T A$?

To compute $A^T A$, we multiply A^T (a 3×2 matrix) by A (a 2×3 matrix). The result is a 3×3 matrix:

$$K = A^T A = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}.$$

How are the entries of K calculated?

Each entry of K is computed as:

k_{ij} = sum of products of corresponding elements in the i -th row of A^T and the j -th column of A .

For example:

- $k_{11} = a_{11}^2 + a_{21}^2$ (sum of squares from the first column of A).
 - $k_{12} = a_{11}a_{12} + a_{21}a_{22}$.
 - $k_{33} = a_{13}^2 + a_{23}^2$.
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Step 3: What Is the Gradient?

The **gradient** tells us how K changes with respect to A . In simpler terms:

- Imagine each element of A as a variable (e.g., $a_{11}, a_{12}, \dots$).
 - The gradient is a matrix that shows how each entry of K depends on each entry of A .
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Step 4: Formula for the Gradient

For the function $K = A^T A$, the gradient of K with respect to A is given by:

$$\frac{\partial K}{\partial A} = 2A.$$

Why?

1. $K = A^T A$, so it involves multiplying A^T and A .
 2. The contribution of each element of A to K is **proportional** to the corresponding element in A . Thus, the derivative results in $2A$.
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Step 5: Why $\frac{\partial K}{\partial A} = 2A$?

Let's verify this step-by-step with a simpler example.

Example:

Suppose A is a 2×2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

1. Compute A^T :

$$A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}.$$

2. Compute $K = A^T A$:

$$K = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Multiply the matrices:

$$K = \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{12} + a_{21}a_{22} \\ a_{12}a_{11} + a_{22}a_{21} & a_{12}^2 + a_{22}^2 \end{bmatrix}.$$

3. Take the derivative of K with respect to A :

- Each entry of K is a sum of terms proportional to A , so the derivative is **twice** A .

Step 6: Final Result

The gradient of $K = A^T A$ with respect to A is:

$$\boxed{\frac{\partial K}{\partial A} = 2A}.$$

Why Is This Important?

This result is useful in many applications (like machine learning and optimization) because it shows how the matrix product $A^T A$ changes with A .